Test Series Code

04

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | DROPPER | **M. MARKS: 720**

DATE: 18/01/2026

SYLLABUS

Physics:	Mechanical Properties of Fluids, Thermal Properties of Matter, Kinetic Theory of Gases, Thermodynamics + 30% Questions from cumulative syllabus of previously conducted all AITS test
Chemistry:	Equilibrium, Electrochemistry + 30% Questions from cumulative syllabus of previously conducted all AITS test
Botany:	Plant Kingdom, Morphology of Flowering Plants + 30% Questions from cumulative syllabus of previously conducted all AITS test
Zoology:	Neural Control & Coordination, Chemical Coordination & Integration + 30% Questions from cumulative syllabus of previously conducted all AITS test

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. A monoatomic ideal gas undergoes a process where the molar heat capacity C varies with temperature T as $C = C_v + \frac{RT}{T_0}$, where T_0 is a constant. The work done by one mole of the gas when its temperature increases from T_0 to $2T_0$ is: ($C_v \rightarrow$ molar heat capacity at constant volume)

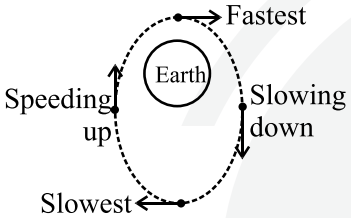
(1) $RT_0 \ln(2)$

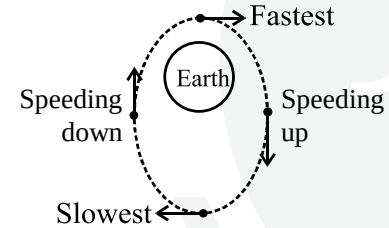
(2) $1.5RT_0$

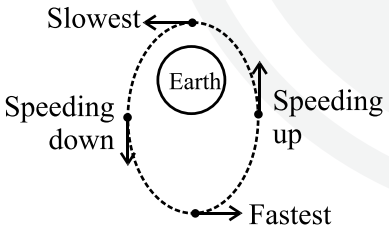
(3) $0.5RT_0$

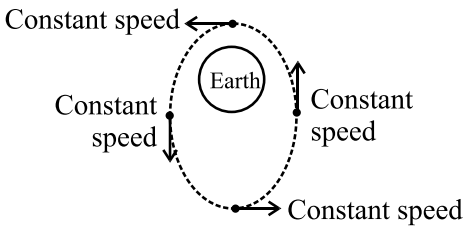
(4) $2RT_0$

2. Choose the diagram which correctly depicts the path of an artificial satellite around earth:

(1) 

(2) 

(3) 

(4) 

3. A vessel contains a mixture of 1 mole of Oxygen gas and 2 moles of Nitrogen gas at $300\ \text{K}$. The ratio of the average rotational kinetic energy per molecule of Oxygen to that of Nitrogen is:

(1) $1:1$

(2) $1:2$

(3) $2:1$

(4) $8:7$

4. A composite rod is made of Copper and Steel joined end to end. The lengths are $1\ \text{m}$ and $2\ \text{m}$ respectively at 0°C. Find the effective coefficient of linear expansion of the composite rod.

($\alpha_c = 1.7\times10^{-5}/^\circ\text{C}$, $\alpha_s = 1.1\times10^{-5}/^\circ\text{C}$)

(1) $1.3\times10^{-5}/^\circ\text{C}$

(2) $1.5\times10^{-5}/^\circ\text{C}$

(3) $1.4\times10^{-5}/^\circ\text{C}$

(4) $1.2\times10^{-5}/^\circ\text{C}$

5. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): A spinning cricket ball in air follows exact same trajectory like a non-spinning ball.

Reason (R): In a spinning ball, according to Bernoulli's theorem, the pressure is higher on the side of the ball where the velocity of air relative to the ball is higher.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) Both A and R are false.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

6. A planet takes 400 days to complete two revolutions around a sun like star in circular orbit. If the distance of the planet from star is reduced to one-fourth of the original distance, then how many days will it take to complete 4 revolutions?

(1) 250 (2) 800

(3) 100 (4) 350

7. Water flows through a horizontal pipe of varying cross-section. At a point where the pressure is P, the velocity is v. At another point where the pressure is $\frac{P}{2}$, the velocity is $2v$. If the density of water is ρ, the value of P is:

(1) $3\rho v^2$

(2) $\frac{2}{3}\rho v^2$

(3) $1.5\rho v^2$

(4) $2\rho v^2$

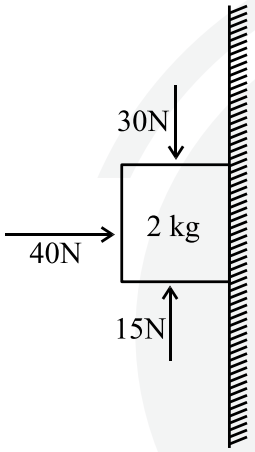
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8. A metal sphere of radius R is heated to temperature T and placed in an enclosure at $0\ \text{K}$. The rate of fall of temperature (by radiation) is proportional to:

(1) R (2) $\frac{1}{R}$

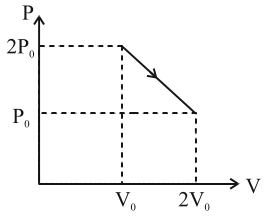
(3) R^2 (4) $\frac{1}{R^2}$

9. A block of mass $2\ \text{kg}$ is pressed against a vertical wall by a horizontal force of $40\ \text{N}$ as shown. Two additional vertical forces of $30\ \text{N}$ and $15\ \text{N}$ are applied to the block. If the coefficient of friction (μ) between the block and the wall is 0.5, what is the magnitude and direction of the frictional force acting on the block? (Take $g = 10\ \text{m/s}^2$)



(1) $35\ \text{N}$, Upward
(2) $20\ \text{N}$, Upward
(3) $35\ \text{N}$, Downward
(4) $20\ \text{N}$, Downward

10. The work done by the gas in the process shown in given *P-V* diagram is:



(1) P_0V_0 (2) $2P_0V_0$

(3) $\frac{3}{2}P_0V_0$ (4) $\frac{P_0V_0}{2}$

11. A small spherical ball falls through a viscous medium with terminal velocity v. If the radius of the ball is doubled, the new terminal velocity will be: (while the other parameters remains same)

(1) v (2) $2v$

(3) $4v$ (4) $8v$

12. The power supplied to a particle of mass 2 kg varies with time as $P = \frac{3t^2}{2}$ Watts. If the velocity of the particle at $t = 0$ is $v = 0$, what is the velocity of the particle at $t = 2$ seconds?

(1) 1 m/s (2) 2 m/s

(3) 4 m/s (4) 8 m/s

13. A copper block of mass $2\ \text{kg}$ is heated to 100°C and placed on a large block of ice at 0°C. The amount of ice that melts is: (Specific heat of copper $= 0.1\ \text{cal/g}^\circ\text{C}$, Latent heat of ice $= 80\ \text{cal/g}$)

(1) 250 g (2) 200 g

(3) 150 g (4) 300 g

14. A body is projected vertically up from the surface of earth with a speed bv_e, where b is a constant ($b < 1$) and v_e is escape speed from the surface of the earth. If R is radius of the earth, then maximum altitude to which the body reaches is;

(1) $\frac{R}{1-b^2}$ (2) $\frac{R}{1-b}$

(3) $\frac{Rb}{1-b^2}$ (4) $\frac{Rb^2}{1-b^2}$

15. A cylindrical tank of height H is filled with water. A small hole is made at a height h from the bottom. The horizontal range of the water jet on the ground is maximum when h equals: (Assume ideal efflux)

(1) $\frac{H}{4}$ (2) $\frac{H}{2}$

(3) $\frac{3H}{4}$ (4) H

16. The degree of freedom of a mixture (at thermal equilibrium) of 1 mole of helium gas and 1 mole of hydrogen gas is: (Neglect vibration)

(1) 3 (2) 5

(3) 4 (4) 4.5

17. The wavelength of maximum intensity of radiation from a star is $500\ \text{nm}$. If the temperature of the star is doubled, the new wavelength of maximum intensity will be:

(1) 1000 nm (2) 250 nm

(3) 500 nm (4) 125 nm

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18. A Carnot engine works between hot reservoir at 600 K and cold reservoir at 300 K. Its efficiency is:

(1) 25% (2) 50%

(3) 75% (4) 100%

19. Given below are two statements:

Statement I: A dimensionless quantity can never have a unit.

Statement II: Unitless quantity must always be dimensionless.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

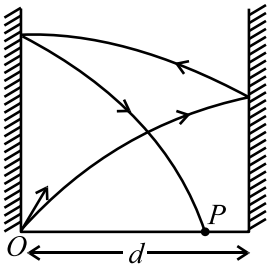
(4) Both Statement I and Statement II are incorrect.

20. The total pressure at the bottom of a lake is 3 times the pressure at the surface. If the atmospheric pressure is 10^5 Pa, the depth of the lake is (Take $g = 10\ \text{m/s}^2$, $\rho = 1000\ \text{kg/m}^3$):

(1) 10 m (2) 20 m

(3) 30 m (4) 40 m

21. A particle is projected from point *O* (the base of a vertical wall) towards another parallel vertical wall located at a distance *d* away. The particle undergoes exactly two elastic collisions with the walls before hitting the ground, as shown in the figure. Find the distance of the landing point *P* from the point of projection *O*. (*R* is the horizontal range of the projectile in the absence of walls, $d < R$)



(1) $R - d$ (2) $2d - R$

(3) $R - 2d$ (4) $3d - R$

22. The root mean square speed of molecules of a gas (H_2) is *v*. If the temperature is quadrupled and the molecules dissociate into atoms, the new rms speed will be:

(1) $2v$ (2) $4v$

(3) $2.82v$ (4) v

23. In a container of fixed volume *V*, there are *N* molecules of an ideal gas, each of mass *m*. If the mass of each molecule is doubled ($2m$) and their root mean square speed is halved $\left(\frac{v_{rms}}{2}\right)$, while the volume and number of molecules remain unchanged, the new pressure P' exerted by the gas will be: (P = initial pressure, V_{rms} = initial rms)

(1) $\frac{P}{2}$ (2) $\frac{P}{4}$

(3) $2P$ (4) P

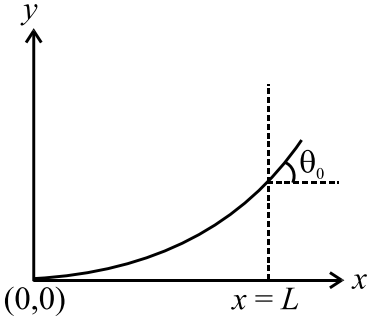
24. An air bubble of radius *R* is formed inside a container of pure water (surface tension *T*). At the same time, a soap bubble of the same radius *R* is blown in the air using a soap solution whose surface tension is $0.5T$. Let P_1 be the excess pressure inside the air bubble in water, and P_2 be the excess pressure inside the soap bubble in air. The ratio $P_1 : P_2$ is:

(1) 1:2 (2) 2:1

(3) 1:1 (4) 4:1

25. Consider a water tank as shown in the figure. It has a vertical wall at $x = L$ and is very wide in the z direction. The tank is filled with a liquid of surface tension S and density ρ. At $x = L$, the liquid surface makes an angle $\theta_0 (\theta_0 << 1)$ with the horizontal. At $x = 0$, the surface is perfectly flat $(y(0) = 0)$. If $\theta(x)$ is the angle that the tangent to the surface makes with the x-axis at any point x, and $y(x)$ is the height of surface, then the expression for $\frac{d\theta}{dx}$ is:

(Take $\theta(x) \approx \sin\theta(x) \approx \tan\theta(x) = \frac{dy}{dx}$, where g is the acceleration due to gravity)



(1) $\frac{d\theta}{dx} = \frac{\rho g}{s} x$ (2) $\frac{d\theta}{dx} = \frac{\rho g}{s} y$

(3) $\frac{d\theta}{dx} = \sqrt{\frac{\rho g}{s}}$ (4) $\frac{d\theta}{dx} = \sqrt{\frac{\rho g}{s}} x$

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26. The mean free path of the molecules of an ideal gas is λ at a certain pressure and temperature. If the gas undergoes an isothermal compression such that its mass density is doubled, the new mean free path of the molecules will be (assume the molecules to be hard spheres of constant diameter):

(1) λ (2) 2λ

(3) $\frac{\lambda}{2}$ (4) 4λ

27. A liquid takes 5 minutes to cool from 80°C to 50°C. How much time will it take to cool from 60°C to 30°C if the room temperature is 20°C?

(1) 5 min (2) 10 min

(3) 9 min (4) 12 min

28. Imagine a hypothetical universe where the force of gravitational attraction F between two bodies is inversely proportional to the cube of the distance r between them. If a planet orbits its star in a stable circular path, how does its orbital time period T vary with the radius r of its orbit?

(1) $T \propto r^{-2}$ (2) $T \propto r^{2}$

(3) $T \propto r^{3/2}$ (4) $T \propto r^{-3/2}$

29. Two small drops of mercury, each of radius r, coalesce to form a single large drop. The ratio of the total surface energy before and after the coalescence is:

(1) $2^{1/3}:1$ (2) $2^{2/3}:1$

(3) $2:1$ (4) $1:2^{1/3}$

30. A black body radiates energy at the rate of E W/m^2 at a high temperature T K. When the temperature is reduced to $\frac{T}{2}$, the new radiant energy will be:

(1) $\frac{E}{2}$ (2) $\frac{E}{4}$

(3) $\frac{E}{8}$ (4) $\frac{E}{16}$

31. A uniform metallic rod of length 100 cm is in a steady state. One end of the rod is maintained at 100°C. The temperature at a point 40 cm from this end is found to be 70°C. What is the temperature of the other end of the rod?

(1) 30°C (2) 10°C

(3) 40°C (4) 25°C

32. A light wire of length 2 m and cross-sectional area 10^{-6} m^2 is suspended from a ceiling. When a mass of 10 kg is hung from it, the elastic potential energy stored in the wire is 0.1 J. What is the Young's Modulus of the material of the wire? ($g = 10$ m/s^2)

(1) 1×10^{11} N/m^2

(2) 2×10^{11} N/m^2

(3) 5×10^{10} N/m^2

(4) 1×10^{10} N/m^2

33. If V is the gravitational potential at the surface of a uniform solid sphere of radius R, at what distance from the center of the sphere is the gravitational potential equal to $\frac{4}{3}$V?

(1) $\frac{3}{4}$R (2) $\frac{4}{3}$R

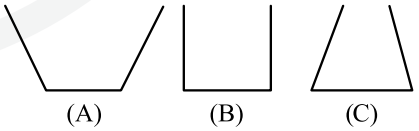
(3) $\frac{R}{\sqrt{3}}$ (4) $\frac{R}{2}$

34. A diatomic gas $(\gamma = 1.4)$ does 200 J of work when it is expanded isobarically. The heat given to the gas is:

(1) 500 J (2) 700 J

(3) 600 J (4) 800 J

35. Three vessels shown in the figure, have same base area. Equal volumes of a liquid are poured in the three vessels. The force on the base due to liquid will be:



(1) Maximum in vessel A

(2) Maximum in vessel B

(3) Maximum in vessel C

(4) Equal in all the vessels

36. A metal rod of length 1 m is clamped at both ends. If the temperature is increased by 10°C, the thermal stress developed is (Young's modulus $Y = 2\times10^{11}$ N/m^2, $\alpha = 10^{-5}/^{o}$C):

(1) 2×10^{7} N/m^2

(2) 2×10^{6} N/m^2

(3) 4×10^{3} N/m^2

(4) 4×10^{7} N/m^2

[Contd...

37. The work done in blowing a soap bubble from a radius of 1 cm to 2 cm is *W*. The work done in blowing it further to radius 3 cm is:

(1) W (2) $1.66W$

(3) $2.33W$ (4) $0.75W$

38. A thin uniform rod of length L and mass M is rotating with an angular velocity ω_0 about a vertical axis passing through its center. Two small beads, each of mass m, are initially at the center and then slide outwards to the ends of the rod (frictionless). The final angular velocity of the system is:

(1) $\frac{M\omega_0}{M+2m}$ (2) $\frac{M\omega_0}{M+6m}$

(3) $\frac{M\omega_0}{M+3m}$ (4) ω_0

39. A satellite of mass *m* is orbiting the earth at a height *h* from its surface. [*M* is mass of the earth and its radius is *R*]

List-I		List-II	
(A)	Kinetic energy of the satellite	(I)	$\frac{GMm}{2(R+h)}$
(B)	Total energy of the satellite	(II)	$\frac{-GMm}{(R+h)}$
(C)	Potential energy of the satellite	(III)	$\frac{-GMm}{2(R+h)}$

(1) A-I, B-III, C-II
(2) A-II, B-III, C-I
(3) A-III, B-I, C-II
(4) A-I, B-II, C-III

40. A car accelerates from rest at a constant rate α for some time, after which it decelerates at a constant rate β to come to rest. If the total time elapsed is *T*, the maximum velocity attained by the car is:

(1) $\frac{\alpha\beta T}{\alpha+\beta}$

(2) $\frac{(\alpha+\beta)T}{\alpha\beta}$

(3) $\frac{\alpha^2 T}{\alpha+\beta}$

(4) $\frac{\beta^2 T}{\alpha+\beta}$

41. In a thermodynamic process, the pressure of a fixed mass of gas is changed in such a manner that the gas releases 20 J of heat and 8 J of work is done on the gas. If the initial internal energy was 30 J, the final internal energy is:

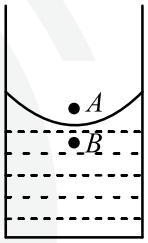
(1) 18 J (2) 42 J

(3) 22 J (4) 58 J

42. A capillary tube is dipped in water and the water rises to height h. If the whole apparatus is taken in an elevator falling freely, the water will:

(1) Not rise at all
(2) Rise to height h
(3) Rise to the top of the tube
(4) Overflow like a fountain

43. A capillary tube is immersed in a liquid, the shape of meniscus is shown below. Choose the **correct** statement.



(1) Pressure at point *A* = Pressure at point *B*
(2) Pressure at point *A* > Pressure at point *B*
(3) Pressure at point *A* < Pressure at point *B*
(4) Data insufficient

44. An ideal monoatomic gas is at an initial pressure *P*. If the gas is compressed adiabatically to $\frac{1}{8}$th of its original volume, the final pressure of the gas will be:

(1) $8P$ (2) $16P$

(3) $32P$ (4) $64P$

45. The ratio of $v_{rms} : v_{mp} : v_{avg}$ for ideal gas is (symbols have their usual meaning)

(1) $\sqrt{3}:\sqrt{2}:\sqrt{\frac{\pi}{8}}$

(2) $\sqrt{3}:\sqrt{2}:\sqrt{\frac{8}{3}}$

(3) $\sqrt{3}:\sqrt{\frac{8}{\pi}}:\sqrt{2}$

(4) $\sqrt{3}:\sqrt{2}:\sqrt{\frac{8}{\pi}}$

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46. Given below are two statements:

Statement I: Conjugate acid of NH_2^- is NH_3.

Statement II: $AlCl_3$ can behave as Lewis acid.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

47. The electrolysis product obtained at cathode during electrolysis of aq. NaCl is:

(1) $H_2(g)$ (2) $O_2(g)$

(3) $Cl_2(g)$ (4) Na(s)

48. Molarity of an aqueous solution of 2% by mass HCl solution having density 1.5 g/ml is:

(1) 0.96 M (2) 0.64 M

(3) 1.24 M (4) 0.82 M

49. The solubility of AgCl in 0.1 M $CaCl_2$ solution will be: (K_{sp} of AgCl = 1.8×10^{-10}):

(1) 9×10^{-10} M (2) 1.8×10^{-9} M

(3) 2×10^{-9} M (4) 1×10^{-11} M

50. Number of Faraday required to oxidise 0.1 mol of I_2 in to IO_3^- is:

(1) 0.5 F (2) 4 F

(3) 2.5 F (4) 1 F

51. The mole of $KMnO_4$ required to oxidise one mol of KI into I_2 in acidic medium is:

(1) 5 (2) $\frac{1}{3}$

(3) $\frac{2}{5}$ (4) $\frac{1}{5}$

52. At a certain temperature and total pressure of 1 atm, chlorine vapours contain 40% by volume of Cl atoms. The equilibrium constant (K_p) for the reaction is:

$Cl_2(g) \rightleftharpoons 2Cl(g)$

(1) 0.66 (2) 1.11

(3) 0.26 (4) 0.88

53. Given below are two statements:

Statement I: For weak electrolytes solution specific conductance decreases on dilution.

Statement II: The electronic conductance decreases with increase of temperature.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

54. The **correct** set of quantum numbers for the last electron of Sodium (Z=11) atom is:

(1) $n = 3, \ell = +2, m = -1, s = +\frac{1}{2}$

(2) $n = 3, \ell = 1, m = -2, s = -\frac{1}{2}$

(3) $n = 2, \ell = 0, m = 0, s = +\frac{1}{2}$

(4) $n = 3, \ell = 0, m = 0, s = -\frac{1}{2}$

55. For which of the following equilibrium reactions at constant temperature on increasing the volume will cause the reaction to shift in forward direction?

(1) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(2) $2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

(3) $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$

(4) $I_2(s) + 5F_2(g) \rightleftharpoons 2IF_5(g)$

56. The **incorrect** statement regarding lead storage battery is:

(1) It is a type of secondary cell

(2) During discharging of battery Pb(s) on anode is converted into $PbSO_4(s)$

(3) A 38% solution of H_2SO_4 is used as an electrolyte

(4) On charging the battery $PbSO_4(s)$ on cathode is converted into Pb(s)

57. The **incorrect** option for a solution of chloroform and acetone is:

(1) $\Delta_{mix}H < 0$ (2) $\Delta_{mix}V < 0$

(3) $\Delta_{mix}G < 0$ (4) $\Delta_{mix}S < 0$

[Contd...

58. Match **List-I** with **List-II**.

List-I (Reaction)		List-II (Unit of K_c)	
(A)	$Fe^{3+}(aq)+3OH^-(aq) \rightleftharpoons Fe(OH)_3(s)$	(I)	Unitless
(B)	$2Cu(NO_3)_2(s) \rightleftharpoons 2CuO(s)+4NO_2(g)+O_2(g)$	(II)	M
(C)	$CaCO_3(s) \rightleftharpoons CaO(s)+CO_2(g)$	(III)	M^{-4}
(D)	$2ICl(g) \rightleftharpoons I_2(g)+Cl_2(g)$	(IV)	M^5

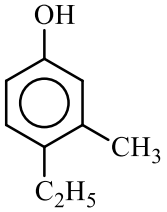
Choose the **correct** answer from the options given below:

(1) A-III; B-IV; C-II; D-I
(2) A-II; B-IV; C-I; D-III
(3) A-IV; B-I; C-II; D-III
(4) A-III; B-I; C-IV; D-II

59. Λ_m^o for $NaNO_3$, HNO_3 and CH_3COONa are x_1, x_2 and x_3 S cm^2 mol^{-1} respectively. The Λ_m^o (in S cm^2 mol^{-1}) for CH_3COOH will be:

(1) $x_2+x_3-x_1$ (2) $x_1+x_2-x_3$
(3) $x_1+x_3-x_2$ (4) $x_1-x_2-x_3$

60. IUPAC name of given compound is:



(1) 4-Ethyl-3-methylphenol
(2) 1-Ethyl-4-hydroxy-2-methylbenzene
(3) 3-Methyl-4-ethylphenol
(4) 4-Ethyl-1-hydroxy-3-methylbenzene

61. The pH of a resulting solution obtained by mixing 100 ml of 0.1 M HCl and 200 ml of 0.2 M HNO_3 will: (Given: log 2 = 0.3010, log 3 = 0.4771)

(1) 0.54 (2) 0.32
(3) 0.78 (4) 0.94

62. Given the standard electrode potentials:

$K^+/K = -2.93\,V$,

$Mg^{2+}/Mg = -2.37\,V$, $Cr^{3+}/Cr = -0.74\,V$

The increasing order of reducing power of the metals is:

(1) K < Mg < Cr
(2) Cr < K < Mg
(3) Mg < K < Cr
(4) Cr < Mg < K

63. The volume of $O_2(g)$ required at STP for complete combustion of 5.8 g of butane will be:

(1) 12.32 L (2) 14.56 L
(3) 16.84 L (4) 20.18 L

64. For the following reactions, equilibrium constants are given:

$A+2B \rightleftharpoons C, K_1$

$2C \rightleftharpoons 2A+4B, K_2$

The **correct** relation between K_1 and K_2 is:

(1) $K_1=\frac{1}{K_2}$ (2) $K_1=\frac{1}{\sqrt{K_2}}$
(3) $K_1=K_2^2$ (4) $K_1=\sqrt{K_2}$

65. A current of 1 A was passed for 965 seconds through an electrolytic cell containing a molten salt of metal X. If 2.5×10^{-3} mol of metal X is deposited at cathode, the oxidation state of X in molten salt is:

(1) +1 (2) +2
(3) +3 (4) +4

66. Oxidation state of sulphur in S_2F_2, SO_2 and S_8 increase in the order:

(1) $S_2F_2 < S_8 < SO_2$ (2) $S_8 < SO_2 < S_2F_2$
(3) $S_8 < S_2F_2 < SO_2$ (4) $SO_2 < S_2F_2 < S_8$

67. The pH of 0.1 M CH_3COONa solution will be:
(Given: pK_a of CH_3COOH = 4.74)

(1) 10.13 (2) 8.87
(3) 5.13 (4) 3.67

68. The standard Gibbs energy of the given reaction is:

$Ni(s)+2Ag^+(aq) \rightarrow Ni^{2+}(aq)+2Ag(s)$

[Given: $E_{cell}^o = 1.05\,V$]

(1) –416.11 kJ mol^{-1} (2) –202.65 kJ mol^{-1}
(3) –336.48 kJ mol^{-1} (4) –184.12 kJ mol^{-1}

69. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: The valence electronic configuration of Cr is $3d^54s^1$.
Reason R: Due to relatively small shielding, smaller coulombic repulsion energy and larger exchange energy, the electronic configuration $3d^5$ has extra stability.
In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

[Contd...

70. In which of the following equilibria, $K_p = K_c$?

(1) $2N_2(g) + O_2(g) \rightleftharpoons 2N_2O(g)$

(2) $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$

(3) $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

(4) $CH_4(g) + 2S_2(g) \rightleftharpoons CS_2(g) + 2H_2S(g)$

71. If the reduction potential of H-electrode at 25°C is –0.413 V, then the pH of solution is:
(Given: $p_{H_2} = 1\,atm$)

(1) 7
(2) 1
(3) 10
(4) 5

72. The standard enthalpy of formation of $CH_4(g)$, $CO_2(g)$ and $H_2O(\ell)$ are x, y and z kJ mol^{-1} respectively. The standard enthalpy of combustion of methane (in kJ mol^{-1}) will be:

(1) x + 2z – y
(2) y + 2z – x
(3) 2y + z – x
(4) y + z – 2x

73. The solubility of $Zn(OH)_2$ in a buffer solution of pH = 12 will be:
(Given: K_{sp} of $Zn(OH)_2$ is 1×10^{-15})

(1) 1×10^{-13} M
(2) 1×10^{-12} M
(3) 1×10^{-14} M
(4) 1×10^{-11} M

74. **Incorrect** statement regarding H_2–O_2 fuel cell among the following is:

(1) the cell was used for providing electrical power in the Apollo space programme
(2) aqueous NaOH is used as an electrolyte
(3) in the cell, energy of combustion of fuel is indirectly converted into electrical energy
(4) pollution free cell

75. Consider the following statements:

A. Solubility of gases in liquids decreases with rise in temperature.
B. Pressure does not have any significant effect on solubility of solids in liquids.
C. 0.9% (mass/volume) NaCl solution is called normal saline solution.
D. Osmosis is used in desalination of sea water.
E. Excess of salty food results a situation called anoxia.

The **correct** statements are:

(1) A, D and E
(2) A, B, D and E
(3) B, C, D and E
(4) A, B and C

76. For the equilibrium,
$NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$,
the equilibrium pressure at a certain temperature is 0.9 atm. The K_p of the reaction is:

(1) 0.108 atm^3 (2) 0.326 atm^3
(3) 0.211 atm^3 (4) 0.432 atm^3

77. The electrical resistance of a column of 0.01 M HCl solution having cross-sectional area 0.1 cm^2 and length 111 cm is 2.22×10^3 ohm. Its molar conductivity will be:

(1) $5 \times 10^4\ S cm^2\ mol^{-1}$
(2) $6 \times 10^5\ S cm^2\ mol^{-1}$
(3) $4 \times 10^3\ S cm^2\ mol^{-1}$
(4) $1 \times 10^6\ S cm^2\ mol^{-1}$

78. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Law of multiple proportions is not applicable for the oxide(s) of magnesium.
Reason R: Magnesium can form only one oxide (MgO).
In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

79. Which pair of electrolytes will show common ion effect?

(1) HCN + NaOH
(2) NH_4Cl + NH_4OH
(3) NaCl + HCl
(4) H_2SO_4 + HNO_3

80. When an external opposite potential of 1.1 V is applied to Daniell cell in standard state, then

(1) Electrons flow from Cu rod to Zn rod
(2) Cu dissolves at cathode
(3) Zn deposits at anode
(4) No flow of electrons

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81. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Disproportionation reaction	(I)	$Cl_2 + 2NaBr \rightarrow 2NaCl + Br_2$
(B)	Non-redox reaction	(II)	$CaCO_3 \rightarrow CaO + CO_2$
(C)	Non-metal displacement reaction	(III)	$CuSO_4 + Zn \rightarrow Cu + ZnSO_4$
(D)	Metal displacement reaction	(IV)	$Cl_2 + 2NaOH \rightarrow NaClO + NaCl + H_2O$

Choose the **correct** answer from the options given below:

(1) A-III; B-IV; C-II; D-I
(2) A-II; B-IV; C-I; D-III
(3) A-IV; B-I; C-II; D-III
(4) A-IV; B-II; C-I; D-III

82. Consider the equilibria:

$2AB(g) \rightleftharpoons A_2(g) + B_2(g)$,

If equilibrium constant is K_p and total pressure at equilibrium is P, then degree of dissociation (α) of AB is given by: (Given: $\alpha <<< 1$)

(1) $\alpha = 2\sqrt{K_p}$ (2) $\alpha = \sqrt{2K_p}$

(3) $\alpha = \sqrt{\frac{2K_p}{P}}$ (4) $\alpha = \sqrt{\frac{2P}{K_p}}$

83. The value of $E^o_{M^{4+}/M^{2+}}$ will be:

(Given: $E^o_{M^{4+}/M^{3+}} = 2.4\ V$ and $E^o_{M^{3+}/M^{2+}} = 0.60\ V$)

(1) 1.2 V (2) 1.5 V
(3) 1.8 V (4) 2.2 V

84. Which of the following **not** increases on increasing the intensity of incident light during photoelectric effect?

(A) Kinetic energy of emitted electrons
(B) Number of photoelectrons
(C) Work function of metal
(1) A and B only
(2) B and C only
(3) A and C only
(4) A, B and C

85. The degree of dissociation of weak base (BOH) having pH = 11 of its 0.1M aq. solution is:

(1) 0.02 (2) 0.12
(3) 0.01 (4) 0.24

86. Given below are two statements:

Statement I: E_{cell} is an intensive property.

Statement II: Electrochemical cells are extensively used for determining the equilibrium constant.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

87. Given below are two statements:

Statement I: Exothermic process are always spontaneous.

Statement II: A system at higher temperature has less randomness in it than one at lower temperature.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

88. On heating a mixture of SO_2Cl_2 and CO, two equilibrium are simultaneously established

$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$

$CO(g) + Cl_2(g) \rightleftharpoons COCl_2(g)$

on adding more SO_2Cl_2 at equilibrium:

(A) Amount of CO will decrease
(B) Amount of $COCl_2$ will increase
(C) Amount of SO_2 will remain unaffected

Choose the **correct** option.

(1) A and B only (2) B and C only
(3) A and C only (4) A, B and C

89. Which among the following ions has maximum Limiting molar conductivity in water at 298 K?

(1) Na^+ (2) Mg^{2+}
(3) K^+ (4) H^+

90. The **correct** option for a galvanic cell is:

(1) $\Delta G < 0, E_{cell} < 0$
(2) $\Delta G < 0, E_{cell} > 0$
(3) $\Delta G > 0, E_{cell} > 0$
(4) $\Delta G > 0, E_{cell} < 0$

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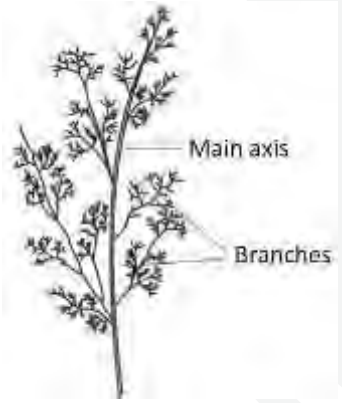
91. Match **List-I** with **List-II.**

List-I (Root Modification)		List-II (Example)	
(A)	For storage of food	(I)	*Rhizophora*
(B)	Prop roots	(II)	Maize
(C)	Stilt roots	(III)	Carrot
(D)	Pneumatophores	(IV)	Banyan tree

Choose the most appropriate answer from the options given below.
(1) A-IV, B-III, C-I, D-II
(2) A-III, B-II, C-IV, D-I
(3) A-I, B-IV, C-II, D-III
(4) A-III, B-IV, C-II, D-I

92. The mycoplasma;
(1) are acellular organisms.
(2) possess cell wall made of pectin.
(3) are members of kingdom Monera.
(4) are obligate aerobes.

93. Identify the given alga and choose the correct option about it.



(1) *Polysiphonia*; stores food in the form of floridean starch.
(2) *Dictyota*; complex post-fertilisation changes are observed during sexual reproduction.
(3) *Porphyra*; thallus possess frond that performs photosynthesis.
(4) *Fucus*; sexual reproduction is of oogamous type.

94. Read the following statements.
A. Biological names are generally in Greek and written in italics.
B. The first word in a biological name represents the lowest category of the taxonomical hierarchy.
C. Both the words in a biological name, when handwritten, are separately underlined.
D. The species *indica* of genus *Mangifera* was first described by Carolus Linnaeus.
E. The name of the author appears before the specific epithet in a scientific name.

In the light of above statements choose the **correct** answer from the options given below:
(1) A, B and C are incorrect, D and E are correct
(2) B, D and E are incorrect, A and C are correct
(3) A, B and E are incorrect, C and D are correct
(4) B, C and D are incorrect, A and E are correct

95. Given below are two statements:
Statement I: Meiosis ensures the production of a haploid phase in the life cycle of sexually reproducing organisms whereas fertilisation restores the diploid phase.
Statement II: The number of bivalents formed during meiosis I is equal to the number of chromosomes in the gamete of that diploid cell.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

96. Identify the **correct** statement.
(1) Theodore Schwann, a botanist, concluded that the presence of cell wall is a unique character of the plant cells.
(2) Matthias Schleiden, a zoologist, observed that all plants are composed of different kinds of cells which form the tissues of the plant.
(3) Antonie Von Leeuwenhoek first saw and described a live cell using an electron microscope.
(4) Rudolf Virchow modified the hypothesis of Schleiden and Schwann to give the cell theory a final shape by explaining that cells arise from pre-existing cells.

97. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In an artificial system of classification, organisms were classified into two groups based on their uses.
Reason (R): Artificial systems separated the closely related species since they were based on a few characteristics.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

98. Read the following statements about members of Phaeophyceae.

A. They are found primarily in marine habitats.
B. The major pigments are chlorophyll *a*, *d* and fucoxanthin.
C. Food is stored as complex carbohydrates in the form of laminarin or mannitol.
D. The vegetative cells have a cellulosic wall usually covered on the outside by a gelatinous coating of algin.
E. Sexual reproduction is by isogamous, anisogamous or oogamous means, and the gametes are pyriform with two apically attached flagella.

The **incorrect** statements are;

(1) B and E. (2) A and D.
(3) C and D. (4) A and B.

99. A flower had 5 petals having keel, standard and wings. Identify the incorrect statement about it.

(1) The corolla is said to have papilionaceous aestivation.
(2) Such flowers are found in bean.
(3) The standard is the largest petal.
(4) The keel overlaps the wings.

100. Identify the incorrectly matched pair.

(1) Psilopsida - *Psilotum*
(2) Lycopsida - *Selaginella*
(3) Sphenopsida - *Pteris*
(4) Pteropsida - *Dryopteris*

101. Given below are some events of a cell cycle.

A. Separation of sister chromatids
B. Formation of chromosomal tetrads
C. Synthesis of genetic material
D. Formation of daughter cells with half the number of chromosomes in each as compared to the parent cell
E. Condensation of chromosomes after interkinesis

Arrange the above events in correct order of occurrence and choose the **correct** option.

(1) A-B-C-D-E (2) C-B-D-E-A
(3) B-E-A-C-D (4) D-A-C-E-B

102. The "anthropocentric view of biology" refers to;

(1) a focus on the systematic and monumental description of all life forms.
(2) the recognition that all present-day living organisms are related to each other.
(3) a perspective that prioritizes human beings and their interests over other forms of life.
(4) a cultural movement for the conservation of biodiversity.

103. Match **List-I** with **List-II**.

List-I (Fungus)		**List-II (Characteristic of concerned class)**	
(A)	*Puccinia*	(I)	Sexual reproduction absent
(B)	*Aspergillus*	(II)	Aseptate and coenocytic mycelium
(C)	*Albugo*	(III)	Asexual spores generally not found
(D)	*Trichoderma*	(IV)	Sexual spores produced in asci

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

104. How many of the following statements about placentation are correct?

A. In marginal placentation, the placenta forms a ridge along the ventral suture of the ovary.
B. In axile placentation, the ovules are attached to placenta in a unilocular ovary.
C. In parietal placentation, the ovules develop at the base of ovary.
D. When the ovules are borne on the central axis and septa are absent, it is called free central placentation.
E. In basal placentation, the placenta develops at the apex of the ovary.

Choose the **correct** option.

(1) Two only (2) Three only
(3) Four only (4) One only

105. Given below are two statements:

Statement I: Malvaceae is commonly called the cotton family.

Statement II: The fruit in members of family Graminae is generally a capsule.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

106. In the seeds of cereals such as maize, the outer covering of endosperm separates the embryo by a proteinous layer called:
(1) scutellum. (2) coleoptile.
(3) coleorhiza. (4) aleurone layer.

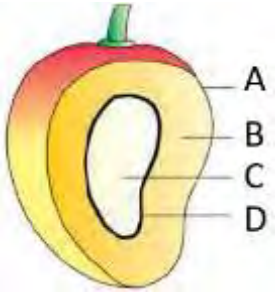
107. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In neem, the rachis that represents the midrib of the leaf bears a number of leaflets.
Reason (R): A bud is present in the axil of leaflets of the compound leaf.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

108. Identify the true statement.
(1) The sporophyte of *Marchantia* consists of foot, seta and capsule but not in *Polytrichum*.
(2) In *Pinus*, the leaves are adapted to increase the surface area.
(3) In *Selaginella*, the development of the zygotes into young embryos take place within the female gametophytes.
(4) *Eucalyptus* is one of the tallest gymnosperms.

109. Identify the characteristics of prokaryotic cells.
A. Well-defined nucleus.
B. Membrane-bound organelles are absent.
C. Ribosomes are 80S type.
D. Specialised differentiated form of cell membrane called mesosome is present.
E. Genetic material is basically naked.
Choose the **correct** option.
(1) A, B and C (2) A, C and E
(3) B, D and E (4) B, C and D

110. Who among the following scientists has been called 'The Darwin of the 20th century' and was awarded the three prizes widely regarded as the triple crown of biology?
(1) Ernst Mayr
(2) R.H. Whittaker
(3) Carolus Linnaeus
(4) G.N. Ramachandran

111. Choose the **correct** statement for the given labels of a mango fruit.



(1) 'A' is not a component of pericarp.
(2) 'D' is a stony structure.
(3) 'C' is the endocarp.
(4) 'B' is fibrous and edible.

112. Identify the **correct** statement about mycorrhiza.
(1) It is an association between a prokaryote and a eukaryote.
(2) One partner is called the mycobiont and another one is called the phycobiont.
(3) It is classified under kingdom Protista by Whittaker.
(4) It is seen in the roots of *Pinus.*

113. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Red algae	(I)	Some of the filamentous forms fix atmospheric N_2 in heterocysts
(B)	Brown algae	(II)	Most members contain pyrenoids in chloroplasts
(C)	Green algae	(III)	Spores or gametes are non-motile
(D)	Blue-green algae	(IV)	Union of gametes may take place in water or inside oogonium

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

114. A heart cell exited the cell cycle after G_1 stage but did not die. Choose the true statement about it.
(1) It has entered the G_2 stage.
(2) It has become dormant and metabolically inactive.
(3) It has entered the quiescent stage.
(4) It can never resume division.

115. Read the following statements regarding bryophytes.

A. The main plant body is haploid and is called a sporophyte.
B. They lack true roots, stem or leaves.
C. The male sex organ is called archegonium.
D. Zygotes undergo meiosis immediately after formation.
E. The sporophyte is not free-living but attached to the photosynthetic gametophyte.

The **correct** statements are;

(1) B and E only. (2) A, B and C.
(3) C, D and E. (4) A and D only.

116. Given below are two statements:

Statement I: In plants such as pea, the seeds are called non-endospermous because the endosperm is present in mature seeds.

Statement II: In a dicotyledonous seed, the hilum is a scar on the seed coat through which the developing seeds were attached to the ovule.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

117. Identify the correct characteristic observed in the members of family Compositae.

(1) Generally, the inflorescence is cymose.
(2) Ovary is one-chambered but it becomes two-chambered due to the formation of the false septum.
(3) Gynoecium is bicarpellary with superior ovary.
(4) Flowers are usually bisexual.

118. Which of the following pairs representing a plant family and its floral formula is correctly matched?

(1) Leguminosae - $\oplus \male \female K_{(5)} \overbrace{C_{1+2+(2)}} A_{(9)+1} \underline{G_{(2)}}$
(2) Solanaceae - $\oplus \male \female K_{(5)} C_{1+2+(2)} A_{(9)+1} \underline{G_{(1)}}$
(3) Brassicaceae - $\oplus \male \female K_{2+2} C_4 A_{2+4} \underline{G_{(2)}}$
(4) Cruciferae - Br (Lemma), Brl (Palea) $\male, \female, P_{2+3}$ lodicules, A_3 or $_{3+3}$, G_1 or $G_{(3)}$

119. In *Spirogyra*, the sexual reproduction occurs by means of;

(1) flagellated and similar sized gametes.
(2) flagellated and dissimilar sized gametes.
(3) non-flagellated and dissimilar sized gametes.
(4) non-flagellated and similar sized gametes.

120. Identify the **correct** statements from the following.

A. The stem tendrils in gourds develop from axillary buds.
B. Thorns in *Citrus* are modified axillary buds.
C. The stem in *Euphorbia* is fleshy and contains chlorophyll.
D. Underground stems in zaminkand are modified to store food.
E. In strawberry, underground stems spread to new niches and form new plants.

Choose the **correct** option.

(1) A, B, C, D only (2) A, B, D, E only
(3) A, B, C, D, E (4) A, C, E only

121. All the following statements are true for numerical taxonomy, except that;

(1) it uses the chemical constituents of the plant to resolve confusions.
(2) it is easily carried out using computers.
(3) number and codes are assigned to all the characters and the data are then processed.
(4) each character is given equal importance and at the same time hundreds of characters can be considered.

122. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Viruses did not find a place in Whittaker's five kingdom classification.

Reason (R): Viruses are obligate parasites that lack genetic material.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

123. Match **List-I** with **List-II**.

List-I		List-II	
(A)	DNA	(I)	Basic proteins present in chromatin
(B)	RNA	(II)	Absent in almost all plant cells
(C)	Histones	(III)	Present in nucleus, mitochondria and chloroplasts
(D)	Centrioles	(IV)	Present in nucleus, cytoplasm, mitochondria and chloroplasts

Choose the most appropriate answer from the options given below.

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-I, C-II, D-III

[Contd...

124. In Pteridophytes, the prothallus;
(1) is large and multicellular diploid structure.
(2) is formed by the germination of spore.
(3) is mostly heterotrophic free-living structure.
(4) represents the dominant phase of life cycle.

125. Which of the following sets of functions are not associated with organelles considered to be a part of endomembrane system?
(1) Glycogen synthesis and steroidal hormone synthesis
(2) Lipid synthesis and glycoproteins formation
(3) Photosynthesis and ATP synthesis
(4) Glycolipid formation and digestion of molecules like carbohydrates and proteins

126. The gemmae in liverworts are;
(1) sexual spores borne on gemma cups.
(2) produced only on female thallus.
(3) unicellular hair-like structures meant for attachment to substrate.
(4) multicellular and green buds.

127. Identify the incorrect statement.
(1) In a flower, the outermost whorl is the accessory organ calyx.
(2) In a floral diagram, the innermost whorl is the reproductive organ gynoecium.
(3) In a flower, radial symmetry means that the flower cannot be divided into two similar halves in any vertical plane passing through the centre.
(4) In a floral formula, fusion is indicated by enclosing the figure within bracket.

128. Given below are two statements:
Statement I: Unlike bryophytes and pteridophytes, in gymnosperms, the male and the female gametophytes have an independent free-living existence.
Statement II: In gymnosperms, the multicellular female gametophyte is retained within the megasporangium.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

129. Identify the correct set of parasitic organisms.
(1) *Trypanosoma*, Tobacco mosaic virus, Venus fly trap
(2) *Plasmodium, Albugo, Cuscuta*
(3) Desmids, Yeast, Bacteriophage
(4) *Euglena, Wolffia, Neurospora*

130. Consider the following statements about Angiosperms:
A. They are an exceptionally large group of plants occurring in a wide range of habitats.
B. The pollen grains and ovules are developed in specialised structures called cones.
C. The seeds are enclosed in fruits.
D. They range in size from the smallest *Wolffia* to tall trees of *Sequoia*.
E. They are divided into two classes: the dicotyledonae and the monocotyledonae.
How many of the above statements are **correct**?
(1) Two only (2) Three only
(3) Four only (4) Five

131. In silk cotton;
(1) pinnately compound leaf is present.
(2) the incisions of a leaf do not reach upto the midrib.
(3) a number of leaflets are present at a common point i.e. the tip of petiole.
(4) leaves lack petiole.

132. In guava,
(1) alternate phyllotaxy is found.
(2) the ovary is said to be inferior.
(3) the flower is said to be hypogynous.
(4) the fruit is always parthenocarpic.

133. Match **List-I** with **List-II**.

List-I		**List-II**	
(A)	Variation in length of stamens	(I)	*Citrus*
(B)	Stamens united in more than two bundles	(II)	Lily
(C)	Stamens attached to the perianth	(III)	Rose
(D)	More than one carpels present freely	(IV)	Mustard

Choose the most appropriate answer from the options given below.
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-I, C-II, D-III

134. Two types of shoots, long and dwarf, are found in:
(1) *Ginkgo*. (2) kelps.
(3) *Equisetum*. (4) *Sphagnum*.

135. If a diploid cell (2n) with DNA content as 2C undergoes mitosis, what will be the chromosome number and DNA content of each daughter cell after the completion of mitosis?
(1) n, C (2) 2n, 4C
(3) n, 2C (4) 2n, 2C

136. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): In a resting neuron, the axonal membrane is comparatively more permeable to potassium ions (K^+) and nearly impermeable to sodium ions (Na^+).

Reason (R): At resting condition, the inner surface of the axonal membrane possesses a positive charge while its outer surface becomes negatively charged.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

137. Read the following statements regarding the mechanism of hormone action:

A. Each receptor is specific to one hormone only and hence receptors are specific.
B. Hormone-Receptor complex formation leads to certain biochemical changes in the target tissue.
C. The hormone-receptor complex in steroid hormones usually interacts with the genome to regulate gene expression.
D. Hormones interacting with membrane-bound receptors do not enter the target cell.

How many of the above statements are **correct**?

(1) One (2) Two
(3) Three (4) Four

138. Given below are two statements.

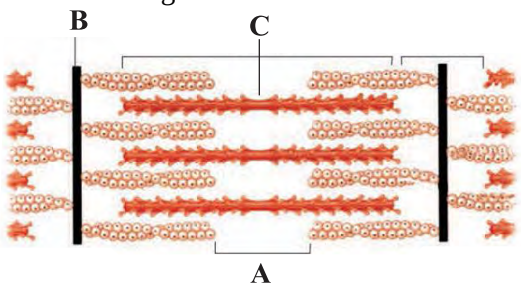
Statement I: The ascending limb of the Loop of Henle is impermeable to water but allows transport of electrolytes actively or passively.

Statement II: The descending limb of the Loop of Henle is permeable to water but almost impermeable to electrolytes.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are incorrect.
(2) Statement I is correct, but Statement II is incorrect.
(3) Statement I is incorrect, but Statement II is correct.
(4) Both Statement I and Statement II are correct.

139. Identify the labelled parts A, B, and C in a sarcomere diagram.



(1) A - H zone, B - M line, C - Actin
(2) A - H zone, B - Z line, C - Myosin
(3) A - M line, B - Z line, C - Actin
(4) A - Z line, B - M line, C - Myosin

140. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Squamous epithelium	(I)	PCT of Nephron
(B)	Cuboidal epithelium	(II)	Bronchioles and Fallopian tubes
(C)	Columnar epithelium	(III)	Walls of blood vessels and air sacs
(D)	Ciliated epithelium	(IV)	Lining of stomach and intestine

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-IV, D-I
(2) A-III, B-I, C-IV, D-II
(3) A-I, B-III, C-IV, D-II
(4) A-II, B-I, C-III, D-IV

141. In the context of Neural Coordination, the electrical potential difference across the resting plasma membrane is maintained largely due to:

(1) Equal efflux of Na^+ and K^+ ions.
(2) Influx of 3 Na^+ and efflux of 2 K^+ ions.
(3) Efflux of 3 Na^+ and influx of 2 K^+ ions.
(4) Negative charge of the myelin sheath.

142. Read the following statements (A-D).

A. The PNS comprises of all the nerves of the body associated with the CNS.
B. The nerve fibres of the PNS are of four types.
C. The functions of the organs/organ systems in our body must be coordinated to maintain homeostasis.
D. Neurons can detect, receive and transmit different kinds of stimuli.

Which of the above statements is/are **incorrect**?

(1) B, C and D only
(2) B only
(3) A and C only
(4) A, B, C and D

[Contd...

143. A patient has a heart rate of 80 beats/minute and a stroke volume of 60 mL. What is the approximate cardiac output and the duration of one cardiac cycle respectively?
(1) 5.8 L/min and 0.8 sec
(2) 4.8 L/min and 0.75 sec
(3) 5.0 L/min and 0.8 sec
(4) 4.8 L/min and 0.6 sec

144. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Glucagon	(I)	Decreases blood pressure
(B)	ANF	(II)	Anti-inflammatory reactions
(C)	Cortisol	(III)	Stimulates glycogenolysis
(D)	Thyrocalcitonin	(IV)	Decreases blood calcium levels

Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-II, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

145. Given below are two statements.
Statement I: In the cockroach, the entire foregut is lined by cuticle.
Statement II: The midgut of the cockroach is broader than the hindgut and is differentiated into ileum, colon and rectum.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct, but Statement II is incorrect.
(3) Statement I is incorrect, but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

146. Read following statements about muscle contraction mechanisms**.**
A. Acetylcholine is released when a neural signal reaches the motor end plate.
B. Muscle contraction begins with a signal from the CNS via a sensory neuron.
C. During muscle contraction, the isotropic band elongates.
D. Repeated muscle activation can lead to lactic acid accumulation.
Select the **correct** statements:
(1) A and D only (2) A and C
(3) B and C (4) A, B, and D

147. Which of the following joints allows limited movement and involves bones joined together with the help of cartilages?
(1) Fibrous joint (Sutures)
(2) Cartilaginous joint (Pubic symphysis)
(3) Gliding joint (Carpals)
(4) Pivot joint (Atlas and Axis)

148. Select the **incorrectly** paired hormone and its associated disease/function:
(1) Growth Hormone – Acromegaly (Hypersecretion in adults)
(2) Vasopressin – Diabetes Mellitus (Hyposecretion)
(3) Thymosin – Differentiation of T-lymphocytes
(4) PTH – Bone resorption (Demineralisation)

149. Given the following respiratory volumes: Tidal Volume = 500 mL, Inspiratory Reserve Volume = 2500 mL, Expiratory Reserve Volume = 1000 mL, Residual Volume = 1200 mL. Calculate the Functional Residual Capacity (FRC).
(1) 2200 mL (2) 3500 mL
(3) 3000 mL (4) 1500 mL

150. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): *Erythroblastosis foetalis* can occur when an Rh-negative mother carries an Rh-positive foetus for the second time (if previously sensitised).
Reason (R): Rh antibodies from the mother can leak into the foetal blood and destroy the foetal RBCs.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

151. Certain aspects of visceral nervous system are given below. Identify the **incorrect** statement.
(1) Visceral nervous system is part of PNS.
(2) Visceral nervous system includes nerve fibres, ganglia and plexus.
(3) Nerves of visceral nervous system travel from CNS to viscera and viscera to CNS.
(4) Visceral nervous system includes only afferent fibres.

[Contd...

152. Select the **incorrect** statement w.r.t. urine formation.
(1) The first step in urine formation is the filtration of blood.
(2) The muscular cells of Bowman's capsule called podocytes are arranged in an intricate manner.
(3) Tubular secretion helps in the maintenance of ionic and acid-base balance of body fluids.
(4) GFR in a healthy individual is approximately 125 mL/min.

153. Regarding the adrenal gland, identifying the **correct** statements:
A. The adrenal medulla secretes catecholamines which are emergency hormones.
B. Zona glomerulosa is the inner most layer of the adrenal cortex.
C. Cortisol is involved in maintaining the cardio-vascular system as well as the kidney functions.
D. Aldosterone acts mainly at the renal tubules and stimulates the excretion of Na^+ and water.
(1) A and C only (2) A, B and C
(3) B and D only (4) A, C and D

154. In male cockroaches, genital pouch lies at the hind end of abdomen bounded dorsally by;
(1) 9th terga and 10th sternum.
(2) 9th sternum and 9th terga.
(3) 9th and 10th terga.
(4) 9th sternum and 10th terga.

155. The sequence of cranial meninges from inner to outside is;
(1) Duramater → Arachnoid → Piamater
(2) Duramater → Piamater → Arachnoid
(3) Arachnoid → Duramater → Piamater
(4) Piamater → Arachnoid → Duramater

156. According to the sliding filament theory of muscle contraction:
A. The 'A' band gets reduced in length.
B. The 'I' band shortens.
C. The 'H' zone disappears or reduces.
D. The Z-lines attached to actins are pulled inwards.
(1) A, B and C only
(2) B, C and D only
(3) A and D only
(4) A, B, C and D

157. Which of the following hormone acts on the exocrine pancreas and stimulates the secretion of water and bicarbonate ions?
(1) Gastrin
(2) Cholecystokinin (CCK)
(3) Secretin
(4) GIP

158. Given below are two statements.
Statement I: A pneumotaxic centre present in the pons region of the brain can moderate the functions of the respiratory rhythm centre.
Statement II: Neural signal from the pneumotaxic centre can increase the duration of inspiration and thereby alter the respiratory rate.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct, but Statement II is incorrect.
(3) Statement I is incorrect, but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

159. Select the **correct** match w.r.t. ABO blood grouping in humans.
(1) Blood group 'A' → Antigen 'A' on RBCs
(2) Blood group 'O' → Anti-A and anti-B antibodies on RBCs
(3) Blood group 'AB' → Donor's group can only AB and B
(4) Blood group 'B' → Donor's group can only B

160. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): Insulin is considered a hypoglycemic hormone.
Reason (R): Insulin acts on hepatocytes and adipocytes to enhance cellular glucose uptake and utilisation, and stimulates glycogenesis.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

161. The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala and hippocampus form a complex structure called:
(1) Corpora quadrigemina
(2) Limbic lobe/system
(3) Corpus callosum
(4) Association area

162. Given below are two statements.
Statement I: The collecting duct allows the passage of small amounts of urea into the medullary interstitium to keep up the osmolarity.
Statement II: Our lungs remove large amounts of CO_2 (approx 200mL/minute) and also significant quantities of water every day.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct, but Statement II is incorrect.
(3) Statement I is incorrect, but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

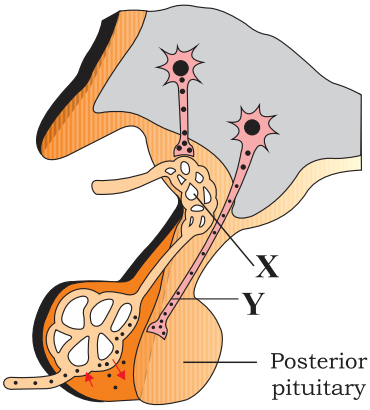
163. Which of the following is a specialised connective tissue that is fluid in nature but does **not** contain fibres?
(1) Cartilage
(2) Bone
(3) Blood
(4) Adipose tissue

164. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Thalamus	(I)	Cardiovascular reflexes and gastric secretions
(B)	Cerebellum	(II)	Major coordinating centre for sensory and motor signaling
(C)	Medulla oblongata	(III)	Body temperature, urge for eating
(D)	Hypothalamus	(IV)	Balance and convoluted surface for neurons

Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-II, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-IV, C-I, D-III

165.



In a diagram of the hypothalamic-pituitary axis, the hormones reaching the Anterior Pituitary are transported via 'X' and hormones reaching the Posterior Pituitary are transported via 'Y'. Identify X and Y.
(1) X - Axonal transport, Y - Portal circulation
(2) X - Portal circulation, Y - Direct neural regulation (Axonal)
(3) X - Lymphatic system, Y - Portal circulation
(4) X - Direct neural regulation, Y - Axonal transport

166. The glenoid cavity articulates the:
(1) Clavicle with the acromion
(2) Scapula with the acromion
(3) Clavicle with the scapula
(4) Head of the humerus with the scapula

167. Consider the statements regarding blood coagulation:
A. It involves a cascade of enzymatic reactions.
B. Thrombin converts inactive Fibrinogen to Fibrin.
C. Thrombokinase is required to convert Prothrombin to Thrombin.
D. Calcium ions play a very important role in clotting.
Which of the above statements are **correct**?
(1) A and B only (2) B and C only
(3) A, C and D only (4) A, B, C and D

168. Choose the **incorrect** statement.
(1) A synapse is formed by the membranes of a pre-synaptic neuron and a post-synaptic neuron.
(2) Hypothalamus is very important part of forebrain.
(3) The electrical potential difference across the resting neural membrane is called the 'resting potential'.
(4) Invertebrates have a more developed neural system.

[Contd...

169. Which of the following hormones play a major role in the regulation of the 24-hour (diurnal) rhythm of our body, such as sleep-wake cycles?
(1) Thymosin (2) Melatonin
(3) Calcitonin (4) Prolactins

170. Select the **correctly** paired respiratory disorder and its cause:
(1) Asthma – Alveolar wall damage
(2) Emphysema – Inflammation of bronchi
(3) Silicosis – Occupational Respiratory Disorder
(4) Asthma – Occupational Respiratory Disorder

171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	PCT	(I)	Concentration of urine (Counter current)
(B)	Loop of Henle	(II)	Conditional reabsorption of Na^+ and water
(C)	DCT	(III)	Reabsorption of 70-80% electrolytes and nutrients
(D)	Bowman's capsule	(IV)	Ultrafiltration

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-IV, C-II, D-I
(4) A-I, B-II, C-III, D-IV

172. In cockroach, a pair of spermatheca is present in the 6th abdominal segment of the female.
Male cockroach has a pair of short thread-like anal styles which are absent in females.
(1) The statement is true for spermatheca but false for anal styles.
(2) The statement is false for spermatheca but true for anal styles.
(3) The statement is true for both spermatheca and anal styles.
(4) The statement is false for both spermatheca and anal styles.

173. Regarding the hormones of the GI tract:
A. Gastrin stimulates secretion of HCl and pepsinogen.
B. CCK acts on the pancreas and gall bladder.
C. GIP stimulates gastric secretion and motility.
D. Endocrine cells present in different parts of the gastro-intestinal tract secrete four major peptide hormones
Which are **correct**?
(1) A, B and D only (2) A and C only
(3) B, C and D only (4) A, B, C and D

174. The gaps between two adjacent myelin sheaths on an axon are called:
(1) Synaptic clefts (2) Nodes of Ranvier
(3) Dendritic spines (4) Schwann gaps

175. Read the following statements regarding Red Muscle Fibres (Aerobic muscles):
A. They possess a high myoglobin content, which gives them a characteristic reddish appearance.
B. They contain a very high number of mitochondria and a high amount of sarcoplasmic reticulum.
C. These fibres primarily depend on aerobic respiration for energy production.
D. Due to the presence of fewer mitochondria, they fatigue quickly during sustained activity.
Choose the **correct** answer from the options given below:
(1) A, B and C only (2) A and C only
(3) B and D only (4) A, C and D only

176. Which of the following statements about Lymph is **false**?
(1) It is a colourless fluid containing specialized lymphocytes.
(2) It acts as a middleman between blood and tissues.
(3) Fats are absorbed through lymph in the lacteals.
(4) It has the same mineral distribution as plasma but possesses more RBCs than blood.

177. Which of the following are generated as Second Messengers when a protein hormone binds to a membrane receptor?
A. Cyclic AMP (cAMP)
B. Calcium ions Ca^{++}
C. IP_3
D. Sodium ions Na^+
(1) A and B only (2) A, B and C only
(3) A and C only (4) A, B, C, D

178. Given below are two statements.
Statement I: In myelinated nerve fibres, the impulse jumps from one node of Ranvier to the next.
Statement II: Conduction is slower in myelinated fibres than conduction in non-myelinated fibres.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct, but Statement II is incorrect.
(3) Statement I is incorrect, but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

179. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): A fall in Glomerular Filtration Rate (GFR) activates the Juxtaglomerular (JG) cells to release Renin.

Reason (R): Renin converts Angiotensinogen in blood to Angiotensin I and further to Angiotensin II, which is a powerful vasodilator.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

180. Identify the **correct** values for partial pressure of CO_2 (pCO_2) in the alveoli and deoxygenated blood respectively.

(1) 40 mm Hg and 45 mm Hg
(2) 45 mm Hg and 40 mm Hg
(3) 104 mm Hg and 40 mm Hg
(4) 40 mm Hg and 95 mm Hg

■■■

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